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Integrating AI and Physics to Mitigate Pre-Mature Failures, Optimize and Stabilize Sulfate Reduction Plant (SRP) in Water Injection Operations

S. A. Ashair, ADNOC Offshore, Abu Dhabi, UAE; R. K Verma, Bosch, Bengaluru, Karnataka, India; H. M. Al Ali, ADNOC Offshore, Abu Dhabi, UAE; R. K. Kothandan, Bosch, Bengaluru, Karnataka, India

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Abstract

The Sulfate Removal Process (SRP) plays a vital role in offshore and onshore oil and gas operations by reducing sulfate concentrations in seawater before it is injected into reservoirs. This process helps prevent scale formation and mitigates reservoir souring caused by sulfate-reducing bacteria. The SRP typically includes a multi-stage filtration system—comprising coarse filtration, ultrafiltration (UF), and nanofiltration (NF). Among these, the UF stage, which targets the removal of fine particulates, has been repeatedly identified as a source of mechanical failure. Specifically plate distortion, bolt rupture, and membrane damage, all occurring under standard operating conditions which lead to reduce production capacity below 70%.

Leveraging the innovative concept of Reality Based Prescriptive Predictive Maintenance (RB-PsM/PdM); a concept innovated by Arthur 1, a combined simulation and AI-driven approach was adopted to diagnose root causes and enable predictive maintenance. Computational Fluid Dynamics (CFD) was employed to analyze internal flow patterns, revealing pressure hotspots and non-uniform load distributions on the membrane. Simultaneously, Finite Element Analysis (FEA) evaluated the structural response of housing components, bolts, and support plates under both filtration and backwash loading. Fatigue life analysis further quantified the long-term impact of cyclic pressurization.

Crucially, insights from these simulations were integrated with operational data using machine learning algorithms to develop a predictive maintenance model. This AI-enhanced model enables near-real-time monitoring and early failure prediction, significantly improving the reliability of the UF system. The study also proposes design upgrades —including an advanced sandwich plate made of Incoloy and Titanium, improved bolt grade and material, and rib-reinforced chamber sections—which collectively reduce cyclic stress concentrations and extend component life.

As a result of the design enhancements and predictive maintenance model, the system's operational reliability significantly improved, leading to a production capacity increase of 12-18%. This translated into extended uptime, reduced unplanned shutdowns, and enhanced overall process efficiency.

The novelty of this work lies in its integrated approach that combines high-fidelity physics-based simulations with artificial intelligence to both diagnose and proactively manage mechanical failures in

critical filtration systems. Unlike conventional failure analysis that is reactive in nature, this study leverages the predictive power of AI trained on insights derived from Computational Fluid Dynamics (CFD), Finite Element Analysis (FEA), and historical operational data. By translating simulation outputs into features for machine learning models, a robust predictive maintenance framework was developed, capable of forecasting component failures—such as bolt loosening, bolt shear, gasket degradation, and membrane rupture—before they occur. This fusion of physics and data-driven intelligence not only enhances the diagnostic depth but also transforms maintenance strategies from time-based to condition-based, paving the way for smarter, more resilient sulfate removal systems in the oil and gas industry.

Introduction

In offshore and onshore oil and gas operations, the quality of injection water plays a decisive role in long-term reservoir performance. Sulfate present in seawater, if not properly removed, can interact with formation minerals to cause scaling and promote microbial activity that leads to souring. To prevent these issues, the Sulfate Removal Process (SRP) employs a multi-stage filtration system—culminating in nanofiltration but critically relying on ultrafiltration (UF) to handle fine particulates. Although UF appears to be a mid-stage operation, it often becomes the focal point of mechanical failures, particularly involving bolts, support plates, and membrane components operating under high-pressure cycles.

Recent operational data from several installations have shown that bolt ruptures, membrane damage, and plate distortions occur even under expected operating ranges. These failures reduce filtration efficiency and force operations to run at under 70% of rated capacity. A closer analysis reveals that these components are subject to cyclic pressurization during filtration and backwash cycles, creating fatigue stress not adequately accounted for during initial design. Moreover, unanticipated flow patterns within the housing units cause uneven pressure loads and local turbulence that further amplify the problem.

To address these recurring failures, this study integrates physics-based simulations with machine learning, forming a comprehensive predictive maintenance framework. The foundation lies in high-fidelity Computational Fluid Dynamics (CFD), which models internal fluid behavior and identifies problematic zones of high shear or stagnation. These flow conditions, once mapped, inform the next step: Finite Element Analysis (FEA). The structural simulations explore how flow-induced loads stress the housing, bolts, and membrane plates under both steady operation and backwash transitions. In parallel, fatigue simulations assess how repeated loading affects the long-term integrity of fasteners and other structural components.

What sets this approach apart is its integration of simulation data with AI algorithms. Insights from CFD and FEA—such as stress contours, deformation zones, and fatigue life—are combined with historical operational data. Machine learning models are then trained on this hybrid dataset to detect patterns and predict potential failures before they occur. This enables a shift from time-based maintenance to condition-based predictive maintenance, minimizing both downtime and maintenance costs.

The study also explores practical design improvements. Material upgrades for bolts and plates, ribbed reinforcement for housing walls, and reconfigured plate geometry are all proposed and validated through simulation. These changes reduce stress concentrations and enhance fatigue resistance, contributing to longer service life and improved performance.

By integrating simulation with real-time analytics, this research advances a scalable and transferable framework for failure prevention in high-pressure filtration systems. The approach goes beyond traditional diagnostics, offering a proactive solution to mechanical reliability issues. It sets a precedent for future digital twin implementations in industrial filtration, making operations not only safer but also more efficient and sustainable.

Literature Survey

The reliability of bolted joints in ultrafiltration (UF) systems, especially in the treatment of seawater, has been a subject of increasing concern due to the unique challenges posed by corrosive marine environments, cyclic hydraulic loading, and complex joint behavior. High-strength bolts subjected to these conditions are prone to fatigue-induced failures that may initiate at thread roots due to geometric stress concentration and are exacerbated by fluctuating mechanical and pressure loads. In a study by Gao et al. (2023), crack propagation in high-strength bolts under fatigue loading was simulated using a combined ABAQUS-FRANC3D methodology. Their approach incorporated detailed thread geometry, initial defects, and preload conditions to simulate crack growth using fracture mechanics. By coupling finite element analysis (FEA) with adaptive meshing and stress intensity factor (SIF) computation, they demonstrated that accurate prediction of bolt life in dynamic environments is possible. Similarly, an experimental study by Lee et al. (2023) analyzed M24 twisted-shear bolts under full preload, revealing how preload enhances fatigue resistance by reducing micromotion. This study provided empirical S-N curve data crucial for accurate fatigue life prediction, especially for bolts under pretension in large-scale process equipment such as seawater filtration modules. Further insights from MDPI's Applied Sciences (2024) showed how environmental temperature plays a critical role in bolt life; tests conducted at 0 °C and 20 °C under preload indicated that low-temperature conditions can improve fatigue performance. These results are particularly relevant to UF systems in marine applications where diurnal or seasonal temperature variations may affect preload retention and fatigue damage accumulation.

In addition to mechanical fatigue, the impact of dynamic loading, such as flow-induced vibration or water hammer effects, must be considered. The Chinese Journal of Mechanical Engineering (2021) explored bolt loosening and fatigue behavior under coupled axial and transverse loads. This study proposed preload optimization as a key strategy to minimize joint failure due to alternating mechanical stresses. Since seawater UF systems often experience dynamic hydraulic conditions—such as pulsation due to pump cycling or membrane backflushing—these findings are highly applicable. CFD (computational fluid dynamics) tools provide a deeper understanding of how hydraulic behavior contributes to bolt failures. An arXiv paper (2022) highlighted how porosity heterogeneity in multibore membrane modules leads to non-uniform flow distribution, generating local turbulence, dead zones, and pressure spikes. When integrated with FEA, CFD data can pinpoint locations of high mechanical stress in bolted flanges, informing structural design improvements. Moreover, a study on marine algal fouling by Zhang et al. (2016) demonstrated how biofilm formation increases transmembrane pressure and disrupts flow uniformity, further increasing mechanical loading on system joints. These hydraulic stressors, if not incorporated into the fatigue analysis of bolted joints, can lead to underestimation of failure risks. The coupling of CFD-derived pressure fluctuations and FEA-based structural response modeling is thus essential for realistic fatigue life assessments in UF systems. Integrating these tools into a co-simulation workflow offers predictive insights into both operational reliability and long-term structural performance of filtration infrastructure. These studies collectively support a methodology where realistic operating loads, temperature variations, and pressure dynamics are captured through multiphysics simulation frameworks.

Methodology

The proposed methodology introduces a novel hybrid digital twin framework for predictive maintenance by combining high-fidelity physics-based models with real-time data analytics. The work began with problem identification in a Sulfate Removal Process (SRP) unit, where frequent bolt failures at the ultrafiltration flange assembly disrupted operations. Initial assessments pointed to fatigue-induced damage caused by cycle flow and pressure cycling (Filtration and Backwash). A comprehensive physics-based modeling approach was developed. First, Computational Fluid Dynamics (CFD) simulations were conducted to analyze the internal flow behavior, revealing pressure surges and flow-induced forces. These results were used as boundary conditions for Finite Element Analysis (FEA), modeling the mechanical response of the bolt-flange assembly under realistic thermal and mechanical loads.

To capture long-term degradation, a fatigue analysis was carried out using the stress-life (S-N) approach, integrating loading spectra from operational conditions. The combined simulation workflow identified high-risk zones and estimated remaining useful life (RUL) of the bolts.

The novelty of the approach lies in the integration of simulation data and operational telemetry into a hybrid digital twin (DT). This twin continuously synchronizes with sensor data (flow rate, pressure, vibration), enhancing predictive accuracy and situational awareness.

The final step involved the development of an interactive cloud dashboard, providing plant engineers with real-time health monitoring, alerts, and decision support tools for proactive maintenance.

This hybrid DT framework offers a replicable template for reliability-centric monitoring of critical equipment across industrial systems.

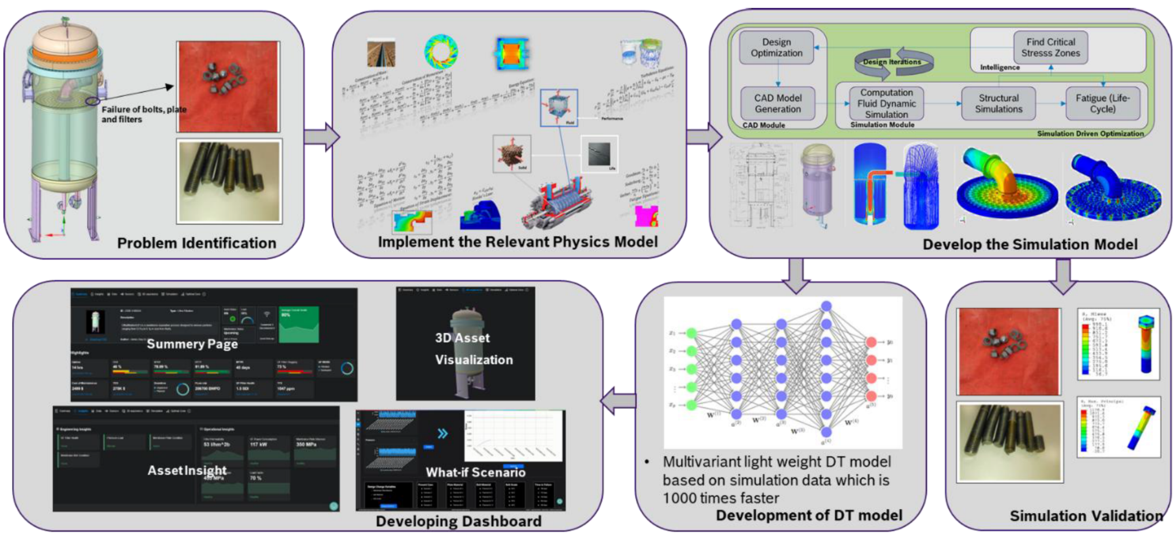

Figure 1—Hybrid digital twin workflow

Simulation

Ultrafiltration (UF) is a membrane filtration process that removes particles between 0.01-0.1 gm, including turbidity, microorganisms, and suspended solids. The system contains 191 membrane bundles mounted on plate that separate filtered and unfiltered water. Pressurized feedwater passes through the membranes, allowing only clean water to permeate. To maintain performance, a backwash operation is periodically triggered, reversing flow to dislodge fouling materials and flush them out. This cleaning ensures sustained efficiency and membrane life. UF is a vital pretreatment step in systems like Sulfate Removal Processes (SRP), protecting downstream equipment by providing high-quality, low-turbidity feedwater.

During filtration and backwash operations, the ultrafiltration membranes experience significant pressure drops across their surfaces. This differential pressure generates substantial forces on the membrane plate that hold the bundles. Membrane plate is subjected to repeated mechanical loading as the pressure cycles between filtration and backwash phases. Membrane plate is secured to the ultrafiltration vessel using 24 M12 bolts, which must withstand the generated loads. Over time, the cyclic pressure and resulting stresses can lead to fatigue or loosening and failure of bolts while operating at rated filtration capacity especially. Ensuring bolt integrity is critical for operational reliability and safety.

Computation Fluid Dynamics (CFD) Analysis

Preprocessing for ultrafiltration simulation involves several critical steps to ensure accurate representation of flow behavior and mechanical loading. First, a detailed 3D CAD model of the ultrafiltration unit is created from engineering drawings, capturing key components such as the membrane plate, housing, filter bundles, and inlet/outlet connections as shown in Fig. 2(a). The internal fluid volume is then extracted from this geometry to define the flow domain required for CFD analysis. Using Ansys Fluent, a high-quality polyhedral mesh with 10 inflation layers is generated, which provides better accuracy and convergence for complex internal flows while keeping the element count manageable as shown in Fig 2(c).

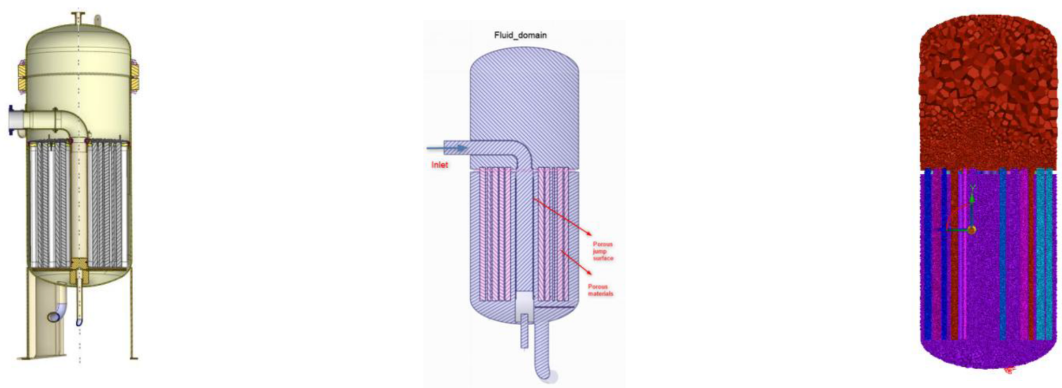

Figure 2—(a) CAD model of the Ultra Filtration, (b) Fluid Model Extraction for the CFD Simulation and (c) Polyhedral mesh generation

Filters are modeled as porous domains, with porosity and directional resistance defined based on manufacturer specified pressure drop data under normal and backwash flow conditions. This approach allows the simulation to realistically predict pressure fields and flow distribution through the membrane bundles. Boundary conditions include specified mass flow at the inlet during filtration and reversed flow for the backwash process, while maintaining pressure outlets or static pressure boundaries at exits. Wall no-slip conditions are applied to solid surfaces. These settings simulate actual operational phases and enable accurate prediction of the pressure drop across the membrane plate, which is later used to calculate mechanical loads for structural and fatigue analysis.

The CFD results of the ultrafiltration system highlight key observations regarding pressure distribution and flow behavior during filtering and backwash operations in in Fig. 3. During the filtering mode, pressure contours show that most of the pressure drop occurs across the membrane filters, which are modeled as porous media. The overall flow distribution appears relatively uniform under standard operating conditions. The uniform pressure drop ensures consistent filtration performance across all membrane bundles.

In contrast, during the backwash operation, where flow is reversed to clean the membranes, the pressure contours reveal a significantly non-uniform pattern. The reverse flow enters through the permeate side and exits via the feed channels, encountering varying resistances across the porous filters. This leads to uneven flow distribution among different bundles, resulting in non-uniform filter cleaning.

Furthermore, the asymmetric backwash flow generates variable pressure gradients across the membrane plate, which is fixed to the ultrafiltration housing by 24 M12 bolts. These non-uniform forces can cause differential stresses on the membrane plate and mounting hardware, increasing the risk of fatigue and failure. The CFD findings underscore the need for improved design and maintenance strategies to mitigate these effects.

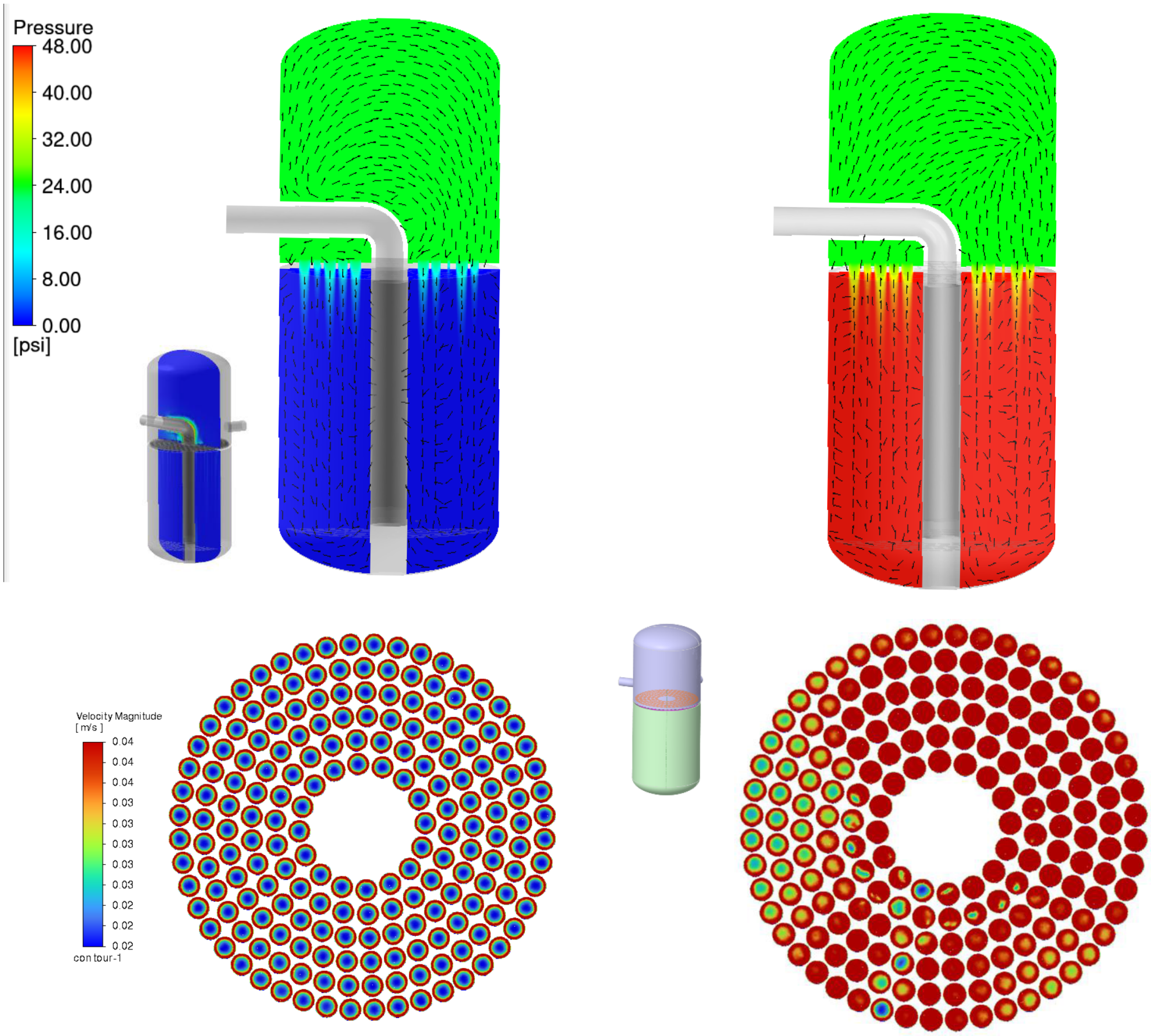

Figure 3—(a) & (b) Pressure contour for filtering and back wash operation, (c) & (d) Velocity distribution at filter entrance for filtering and backwash operation

Finite Element Analysis (FEA) Simulation

The pressure distribution obtained from the CFD simulations for both filtering and backwash operations is mapped as boundary loading conditions onto the structural model of the membrane plate in Abaqus. These pressure loads are applied on the surfaces where the membrane filters interact with the plate. Separate FEA simulations are conducted for both operating conditions to assess the resulting stress and deformation patterns. The filtering operation shows uniform loading, while the backwash operation induces asymmetric forces, leading to uneven stress concentrations. These results help identify critical regions around bolt holes and plate edges that are susceptible to fatigue and potential failure.

The FEA simulations reveal distinct stress patterns in the membrane plate and bolts under filtering and backwash operations. During the filtering process, the pressure acts upward on the membrane plate, causing the plate to deform upwards. This upward displacement creates a bending effect at the bolt head region, as the plate tends to lift while the bolts restrain its motion. As a result, high tensile and bending stresses are concentrated at the bolt head and bolt areas towards the membrane plate, particularly near the bolt holes as shown in fig. 4.

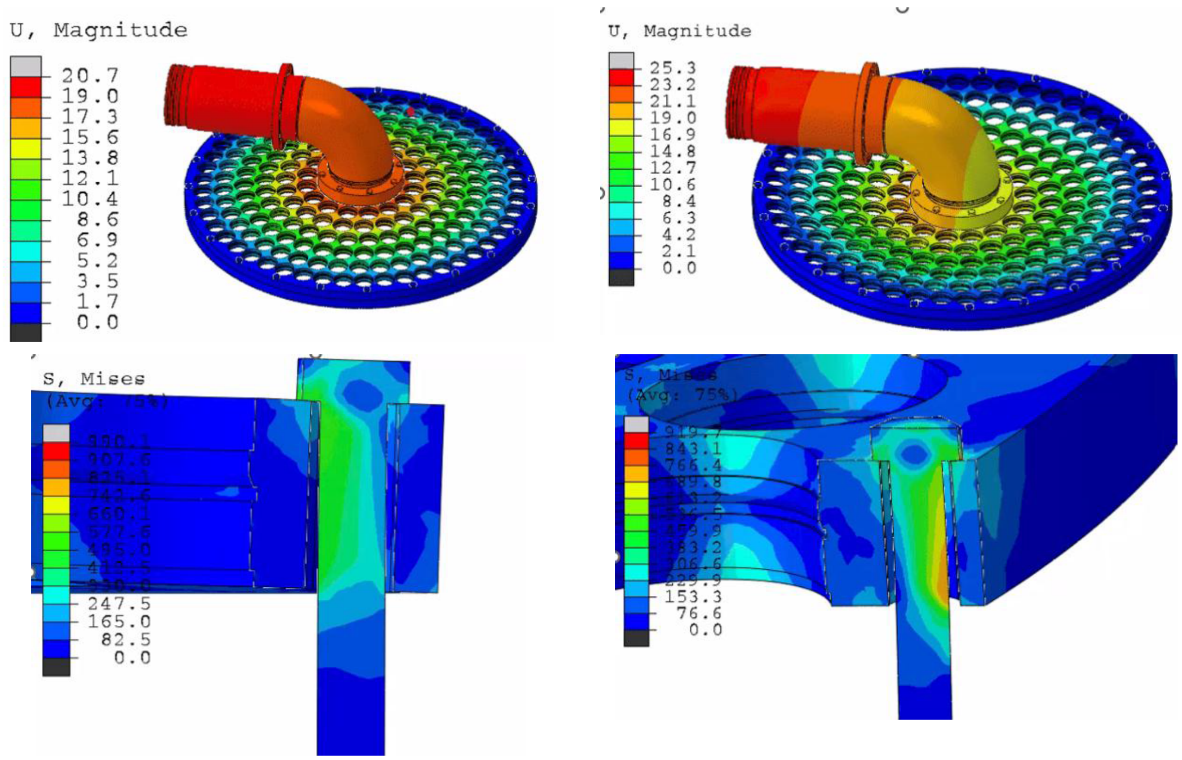

Figure 4—(a) & (b) Membrane plate assembly deflection contour for filtering and backwash operation, (c) & (d) Bolt Von-mises stresses contour for filtering and backwash operation

In contrast, during the backwash operation, the flow is reversed, and the pressure acts downward on the membrane plate. This causes the plate to deflect downward, and the bolts experience significant bending other side of the bolt. This downward motion induces a different loading condition, where the stress is concentrated around the bolt threads and shank. The analysis indicates that the maximum stress locations in the bolt vary between the two operating modes.

These alternating high-stress regions pose a fatigue risk to the bolt and plate interface. The non-uniform stress response between the filtering and cleaning cycles highlights the importance of considering both loading scenarios during design and failure analysis to ensure long-term structural integrity.

The FEA results indicate that the membrane plate experiences significant localized stresses as shown in fig.5. The thickness of the plate in these areas is relatively small, leading to stress concentration during both filtering and backwash operations. Under pressure loading, the reduced cross-sectional area between holes cannot effectively distribute the applied loads, resulting in elevated stress levels. These narrow sections act as weak zones where bending and tensile stresses accumulate. Over time, such stress concentrations initiate cracks or permanent deformation, compromising the structural integrity of the membrane plate and contributing to premature failure.

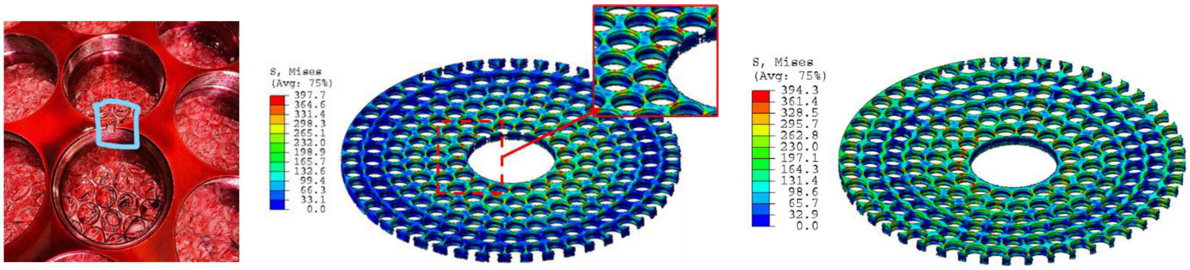

Figure 5—(a) Actual membrane plate failure images, (b) & (c) Von-mises stresses contour over bolt for filtering and backwash operation

Fatigue Life Assessment

In this study, fatigue analysis was performed using the Abaqus simulation tool to estimate the life of bolts and membrane plate under cyclic loading caused by repeated filtering and backwash operations. Abaqus uses stress-life (S-N) based approaches for fatigue evaluation, especially suitable for cyclic fatigue conditions typically seen in such systems. The fatigue tool in Abaqus evaluates damage based on signed von Mises equivalent stress amplitude and mean stress correction, often using Goodman or Gerber criteria for adjusting the effects of mean stresses.

In the Abaqus fatigue module, the tool integrates stress history data from the FEA results and applies it over defined loading cycles to compute fatigue damage per cycle. Cumulative damage is then evaluated using Miner's Rule, which adds the ratio of applied cycles to failure cycles at different stress levels. A damage value of 1 indicates failure. This approach helps identify critical regions with potential fatigue failure, such as bolt threads and thin sections of the membrane plate.

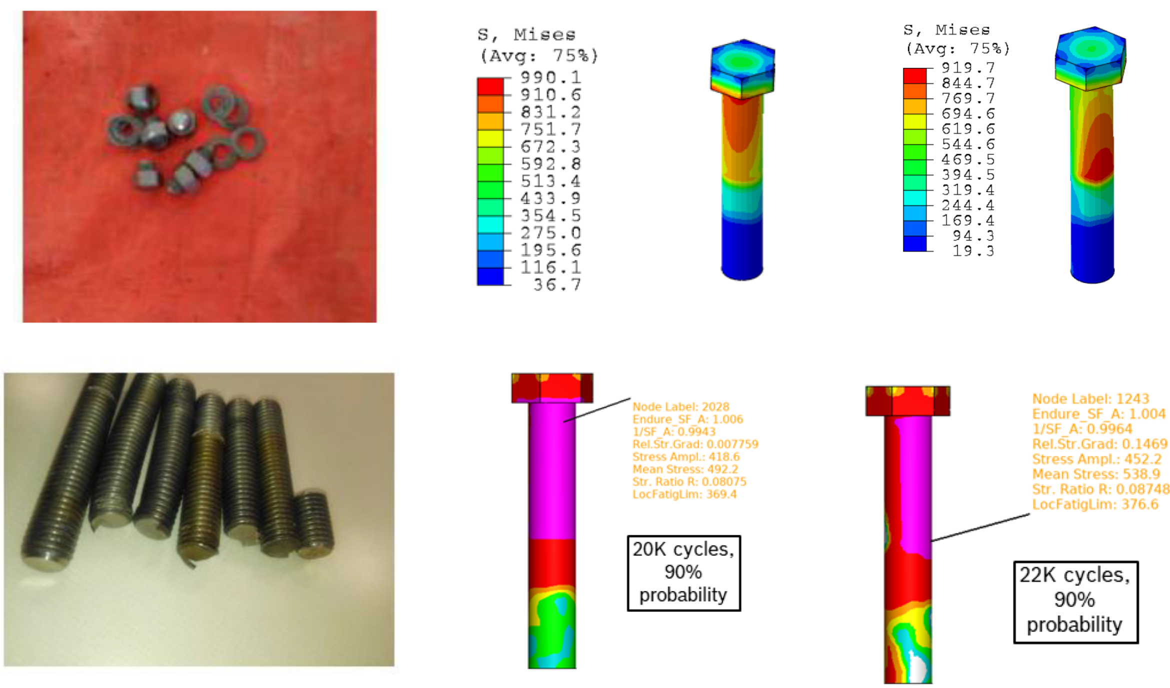

Figure 6—(a) & (d) Actual bolt failure images, (b) & (c) Bolt Von-mises stresses contour over bolt for filtering and backwash operation, (e) & (f) Fatigue life assessment contour for filtering and backwash operation

Hybrid Digital Twin (Confidential)

A hybrid digital twin model is developed to enhance the predictive capability. Data science-based surrogate model was developed using simulation data generated under a range of operating conditions.

A comprehensive set of CFD and FEA simulations were conducted for varying inlet pressures and flow rates, replicating different real-world scenarios for both filtering and backwash operations. Each simulation produced detailed results including pressure distributions, force profiles on the membrane plate, stress maps on bolts, and fatigue damage estimates using stress-life fatigue theory.

The simulation data, comprising inputs (operating pressure and flow) and outputs (maximum stress, fatigue life, damage factor), served as the training dataset for the development of AI models. Various machine learning algorithms—such as Random Forest, Support Vector Regression (SVR), Gradient Boosting, and Deep Neural Networks—were evaluated for their ability to accurately learn the non-linear relationships between operating conditions and structural responses.

To transition from simulation-based to real-world deployment, the trained model was integrated with historical sensor data collected from the ultrafiltration system. This sensor data included flow rates, pressures, cycle durations, and operating frequencies over time. The AI model consumed this time-series data to estimate stress levels and damage accumulation during each filtration and backwash cycle.

Using Miner's Rule, the model continuously evaluated the cumulative fatigue damage on each bolt throughout its operational history. This enabled prediction of remaining useful life (RUL) for each bolt under current operating trends. Notably, the model predicted bolt failure in alignment with the actual failure timeline observed in the field, demonstrating its robustness and reliability.

By combining physics-based simulation with AI-driven inference, the hybrid data science model adds predictive intelligence to the digital twin. This empowers maintenance teams with early warnings, condition-based alerts, and accurate life predictions. The model serves as a foundation for real-time digital monitoring, smart maintenance scheduling, and overall improvement in system reliability and safety. Future enhancements may include online learning, anomaly detection, and adaptive control integration.

Results and Discussion

The hybrid digital twin developed for the ultrafiltration system provided deep insights into the failure mechanisms of the membrane plate and bolts. The CFD simulations revealed that the majority of the pressure drop occurs across the membrane filters during both filtering and backwash operations. This generated significant pressure differentials across the membrane plate, leading to substantial fluid-structure interaction forces.

The CFD results indicated that during the filtering process, the flow pushes the membrane plate upwards, whereas during backwash, the flow direction reverses, causing the plate to be forced downward. This alternating flow behavior results in non-uniform pressure distributions, particularly during the backwash phase, which contributes to non-uniform clogging of filters and asymmetric force loading on the plate.

These force data were used as input for FEA simulations in Abaqus. The FEA results showed that during filtering, the bolt head area experiences high bending stress due to the plate's upward motion. In contrast, during the backwash cycle, the stress is concentrated near the threaded region of the bolt as the plate is pulled downward. The membrane plate itself was found to experience high stress concentrations between bolt holes due to insufficient thickness, creating zones of potential cracking and fatigue failure.

The fatigue analysis, conducted using stress-life (S-N) methods in Abaqus, incorporated the alternating stresses from cyclic operation and applied Miner's Rule to evaluate cumulative fatigue damage. The predicted fatigue life for the bolts under actual operational conditions was approximately 1 year, which closely matched the observed failure timeline in the field. Physical inspection confirmed that the actual bolt failures corresponded to the high-stress regions identified in the FEA simulation—namely, the head and threaded regions. This strong correlation between simulated stress regions, fatigue life predictions, and actual failure observations validates the hybrid digital twin methodology. It demonstrates the value of integrating physics-based modeling (CFD + FEA) with operational data and fatigue theory to diagnose and predict component failures in complex water treatment systems.

The proposed digital twin framework not only helps in root cause analysis of existing failures but also provides a predictive capability to support proactive maintenance planning. Future extensions can include real-time monitoring, adaptive control strategies, and integration with asset management platforms to enhance reliability and reduce downtime in critical infrastructure.

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